

TAE-I
Project Based Learning
on
PCA Based Feature Reduction
Submitted By
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Under the Course
Pattern Recognition- (N-PECCM703T)
B.Tech VII Semester
Academic Session 2026-27(ODD)

Under the Guidance
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Aim: The aim of this project is to develop an application that uses Principal Component Analysis (PCA) to reduce the number of dimensions in Spotify music data while preserving most of its important information. The reduced data is then visualized to make patterns and relationships between different music tracks easier to understand. As an additional analysis, K-Means clustering is applied to the PCA-reduced data to identify groups of tracks having similar audio characteristics.

Objectives:

The main objectives of the project are:

1. To explore a large Spotify tracks dataset containing different audio characteristics.
2. To clean the dataset and remove unnecessary or duplicate records.
3. To select meaningful numerical audio features for analysis.
4. To standardize the selected features before applying PCA.
5. To understand the working of Principal Component Analysis.
6. To reduce the dimensionality of the Spotify audio-feature dataset using PCA.
7. To determine the number of principal components required to retain most of the information.
8. To study the contribution of original audio features to each principal component.
9. To visualize the reduced data using principal components.
10. To apply K-Means clustering to the reduced feature space for additional pattern analysis.
11. To build an application that brings the preprocessing, PCA analysis, visualization, and clustering results together.

Introduction:

Music streaming platforms generate a large amount of data about every track. Along with information such as artist, album, genre, and popularity, Spotify provides several numerical audio features that describe the characteristics of a song.

Some of these features include danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

Although these features are useful individually, studying all of them together becomes difficult when the dataset contains thousands of tracks. It is also not possible to directly visualize a dataset containing nine or more dimensions on a normal two-dimensional graph.

This project addresses this problem using Principal Component Analysis (PCA). PCA converts the original features into a smaller set of new variables called principal components. These components are created in such a way that the first component captures the maximum possible variation in the data, the second captures the next highest amount, and so on.

The project then uses these components to visualize the Spotify dataset. K-Means clustering is also applied after PCA to explore whether tracks with similar characteristics naturally form groups.

Therefore, the overall focus of the project is:

Reducing complex Spotify audio data using PCA and making the resulting patterns easier to visualize and understand.

Problem / Goal:

The Spotify dataset contains a large number of songs, and each song is described using several numerical audio features such as danceability, energy, loudness, speechiness, acousticness, instrumentality, liveness, valence, and tempo.

Although these features provide useful information about music, analyzing all of them simultaneously becomes difficult when trying to understand relationships and patterns visually. With nine dimensions, it is not possible to directly visualize the complete feature space in a simple two-dimensional graph.

Another challenge is that some audio features may contain overlapping or correlated information. Working with all features independently can therefore make analysis more complicated than necessary.

Goal of the Project

The main goal of this project is to develop a **PCA-based feature reduction and visualization system** that converts the original nine-dimensional Spotify audio-feature space into a smaller number of meaningful principal components while preserving most of the information contained in the original data.

The project also uses **K-Means clustering as a supporting analysis technique** to explore whether songs form groups with similar audio characteristics after dimensionality reduction.

The main goal can therefore be summarized as:

Reduce the dimensionality of Spotify audio features using PCA, preserve the important information, and make the resulting patterns easier to visualize and analyze.

Solution:

The proposed solution is a **PCA-based music feature reduction and visualization system** developed using the Spotify Tracks Dataset.

Instead of analyzing all nine audio features independently, the system first cleans and prepares the data, standardizes the selected features, and then applies Principal Component Analysis (PCA).

The PCA transformation produces new variables called **principal components**. These components are ordered according to the amount of variance they explain. The most important components can then be retained while removing less important dimensions.

For this project, PCA was applied to the following nine features:

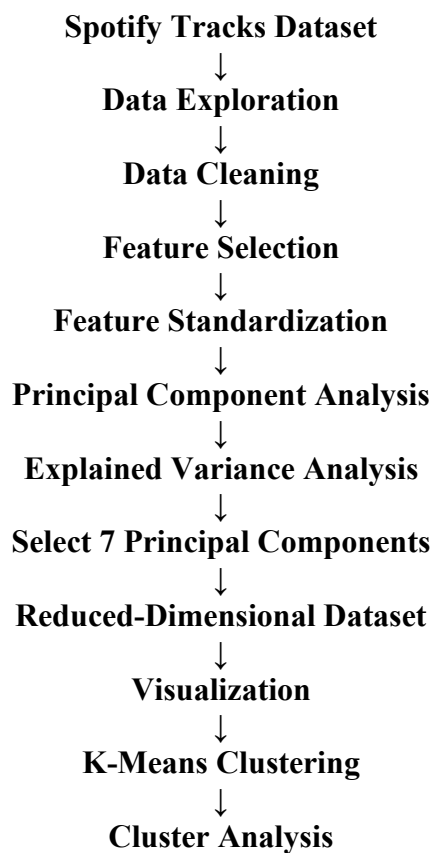
- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentality

- Liveness
- Valence
- Tempo

After analysis, **7 principal components were retained**, preserving approximately **94.81% of the total variance**.

The reduced representation is then used for visualization and further analysis. K-Means clustering is applied to the seven-dimensional PCA representation to identify groups of tracks with similar audio characteristics.

Proposed Solution Pipeline



The important distinction is that **PCA is the primary technique of the project**, while K-Means is used as an additional analytical step.

Dataset Description

The project uses the **Spotify Tracks Dataset**, containing approximately **114,000 music tracks** and **20 attributes**.

The dataset contains information about songs, artists, albums, genres, popularity, and numerical audio characteristics.

For PCA, nine numerical audio features were selected because they directly describe musical characteristics.

The original dataset contained:

114,000 rows × 20 columns

After removing repeated track_id entries, the dataset used for the main PCA analysis contained:

89,741 unique tracks

Thus:

114,000 → 89,741 tracks

The reduction was performed at the track-ID level because repeated IDs contained identical audio-feature vectors, making them redundant for PCA analysis.

Theoretical Background

Dimensionality Reduction

Dimensionality reduction is the process of representing data using fewer variables while attempting to preserve the important information present in the original dataset.

High-dimensional data can create several difficulties:

- Difficult visualization
- Increased computational complexity
- Redundant information
- Difficulty identifying important patterns
- More complicated analysis

Dimensionality reduction techniques transform the original feature space into a smaller representation.

One of the most widely used techniques for this purpose is **Principal Component Analysis (PCA)**.

Principal Component Analysis

Principal Component Analysis is a statistical and mathematical technique used to transform a dataset containing potentially correlated variables into a new set of uncorrelated variables called **principal components**.

The principal components are arranged according to the amount of variance they explain.

The first principal component explains the largest possible amount of variance.

The second principal component explains the largest remaining amount of variance while being orthogonal to the first component.

This process continues for the remaining components.

In simple terms:

PCA changes the coordinate system of the data so that the most important patterns are represented by the first few dimensions.

PCA Processing Flowchart

Figure 1. PCA Processing Flowchart



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Figure 1. PCA Processing Flowchart

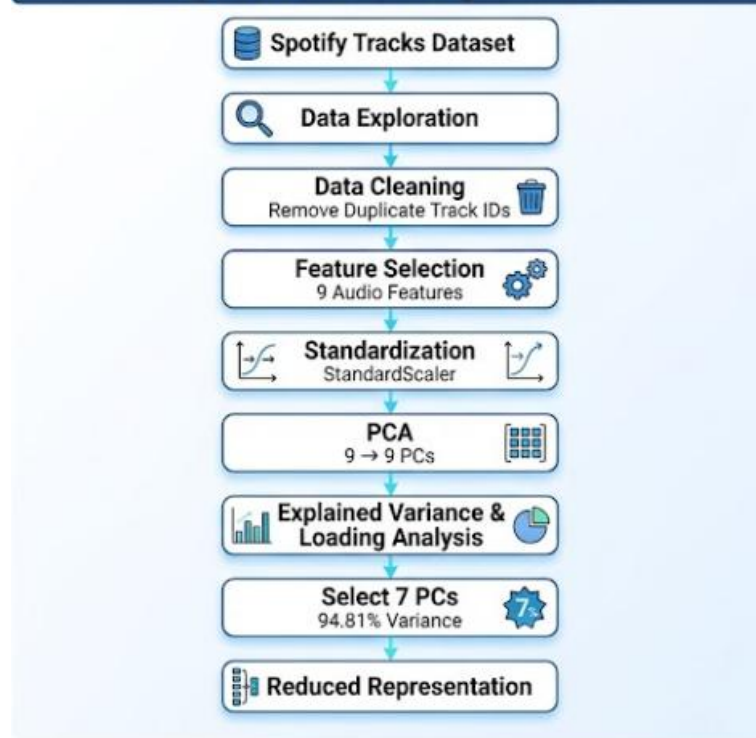
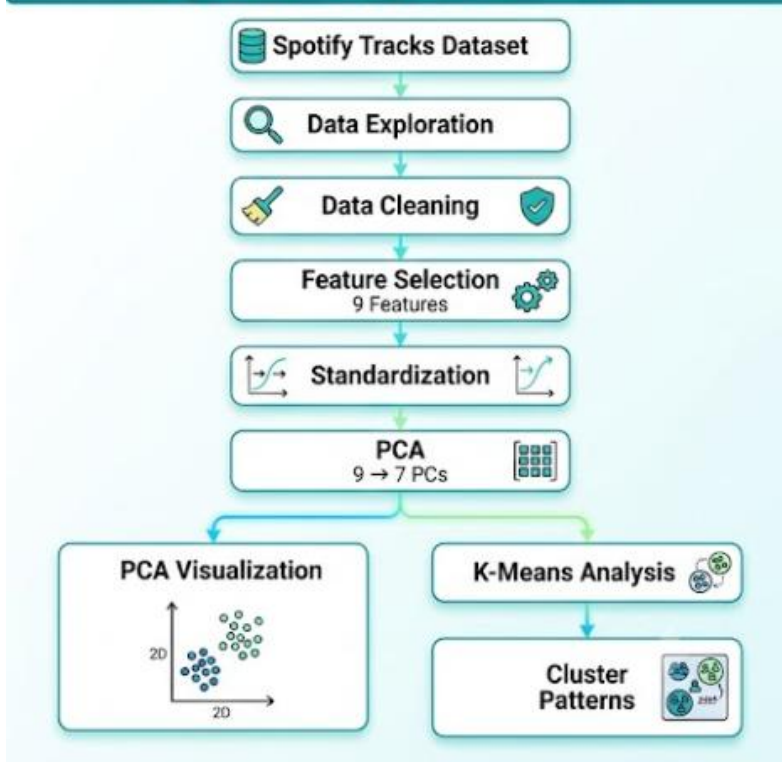


Figure 2. Overall System Architecture



PCA Methodology

The PCA implementation follows these steps.

Step 1 — Data Exploration

The dataset was inspected to understand:

- Number of rows and columns
- Data types
- Missing values
- Duplicate records
- Distribution of numerical features

Step 2 — Data Cleaning

Duplicate track IDs were identified and removed.

The cleaned dataset contained **89,741 unique tracks**.

Step 3 — Feature Selection

Nine audio features were selected for PCA.

Metadata such as artist, album, and track name was not used as PCA input because these fields are categorical or descriptive rather than continuous numerical audio measurements.

Step 4 — Standardization

The selected features were standardized using StandardScaler.

This ensures that features with larger numerical scales do not dominate the PCA calculation.

Step 5 — PCA Transformation

PCA was initially fitted using all nine standardized features.

This produced nine principal components.

Step 6 — Explained Variance Analysis

The explained variance of each component was calculated to determine how much information each component retains.

Step 7 — Component Selection

Seven principal components were selected because they retain approximately **94.81%** of the total variance.

PCA Results and Analysis

After preprocessing and standardization, PCA was applied to the nine selected Spotify audio features: danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

The PCA results show that the first principal component explains **32.12%** of the total variance, while the second and third components explain **15.87%** and **13.89%**, respectively. Together, the first three components preserve approximately **61.88%** of the information contained in the original feature space.

A major objective of dimensionality reduction was to determine how many principal components were required to retain most of the original information. The cumulative explained variance reaches **89.74% with six components** and **94.81% with seven components**.

Therefore, seven principal components were selected for the reduced representation.

Table 1. Explained Variance by Principal Component

Component	Variance Explained	Cumulative Variance
PC1	32.12%	32.12%
PC2	15.87%	47.99%
PC3	13.89%	61.88%
PC4	10.00%	71.88%
PC5	9.69%	81.57%
PC6	8.17%	89.74%
PC7	5.07%	94.81%
PC8	3.63%	98.44%
PC9	1.56%	100.00%

Thus, the original nine-dimensional feature space can be represented using seven principal components while retaining approximately **94.81% of the total variance**.

PCA Visualization and Feature Interpretation

Although seven components were retained for the reduced representation, the first two principal components are useful for visualization because they provide a two-dimensional representation of the dataset.

PC1 explains **32.12%** of the variance and has relatively strong positive contributions from **loudness and energy**, while acousticness contributes in the opposite direction. This component can therefore be broadly interpreted as an **energy/intensity-related dimension**.

PC2 explains **15.87%** of the variance. It has strong positive contributions from **danceability and valence**, making it broadly associated with characteristics related to **danceability and musical mood**.

PC3 explains **13.89%** of the variance and is strongly influenced by **liveness and speechiness**, indicating a dimension related to live performance and spoken-content characteristics.

It is important to note that PCA component signs are mathematically arbitrary. Therefore, the interpretation should focus mainly on the **relative magnitude and direction of the feature loadings** rather than treating the sign as an absolute meaning.



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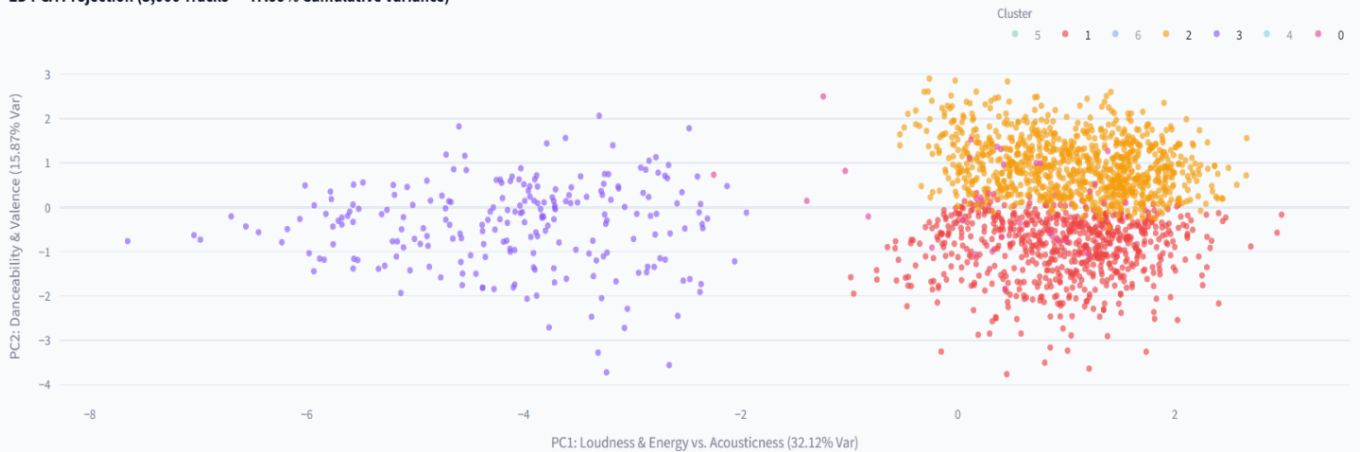
Table 2. Important PCA Feature Loadings

Feature	PC1	PC2	PC3
Danceability	0.258	0.562	-0.093
Energy	0.505	-0.289	0.001
Loudness	0.515	-0.081	-0.049
Speechiness	0.097	0.032	0.641
Acousticness	-0.433	0.315	0.217
Instrumentalness	-0.283	-0.314	-0.213
Liveness	0.084	-0.231	0.673
Valence	0.299	0.527	0.004
Tempo	0.202	-0.251	-0.183

PC1–PC2 Visualization

The PC1–PC2 projection provides a visual representation of the tracks after dimensionality reduction. The two axes together represent approximately **47.99% of the total variance**.

2D PCA Projection (3,000 Tracks — 47.99% Cumulative Variance)





The visualization helps demonstrate how tracks are distributed within the reduced feature space. Tracks with similar combinations of audio characteristics tend to appear closer together, while tracks with substantially different characteristics appear farther apart.

K-Means Clustering on PCA Features

After dimensionality reduction, K-Means clustering was applied to the **seven retained principal components**. The purpose of this step was not to replace PCA, but to use the reduced representation to identify groups of tracks with similar audio characteristics.

PCA and K-Means therefore perform two different tasks:

PCA → reduces the dimensionality of the data.

K-Means → groups similar observations into clusters.

Using the seven PCA components instead of the original nine audio features also provides a compact representation for clustering while retaining approximately 94.81% of the original variance.

Different values of K were evaluated from **2 to 10**. The analysis used a fixed random state of 42 and multiple initializations to improve the stability of the K-Means solution.



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Table 3. K-Means Evaluation

K	Silhouette Score	Inertia
2	0.2720	591132.89
3	0.1759	520026.42
4	0.1845	463963.26
5	0.1903	415332.84
6	0.2030	373839.45
7	0.2108	337715.45
8	0.1909	317238.14
9	0.1984	298832.81
10	0.1844	285040.40

The inertia decreases as K increases, which is expected because more clusters generally reduce within-cluster variation. However, the improvement becomes progressively smaller at higher values of K.

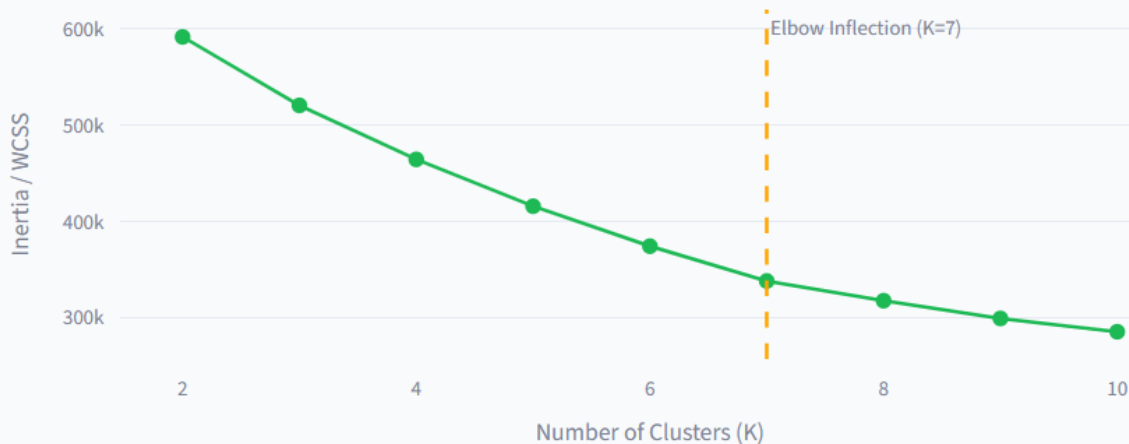
The silhouette score is highest at K=2, but K=2 provides only a very broad division of the dataset. Among the multi-cluster solutions, **K=7 provides the strongest silhouette behavior while also producing meaningful and interpretable groups.**

Therefore:

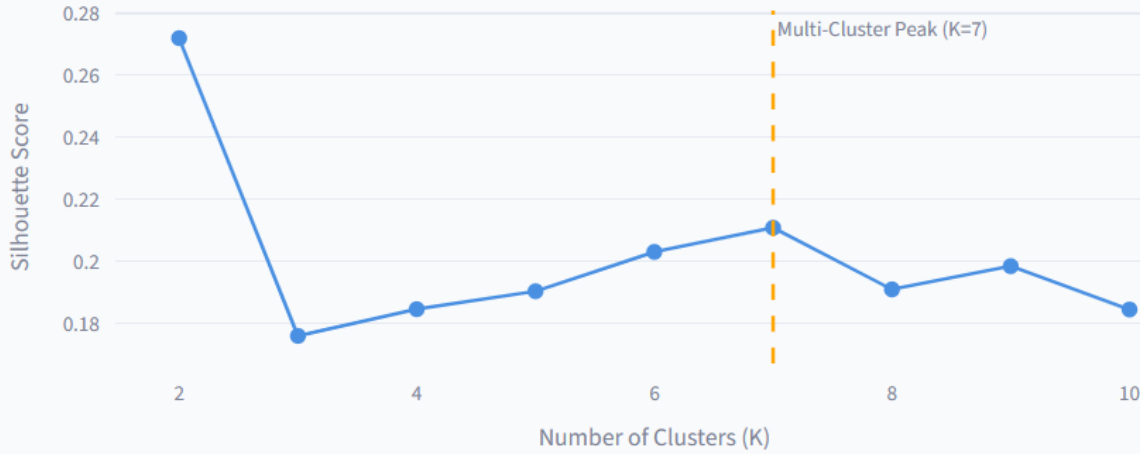
K=7 was selected as the final practical multi-cluster solution based on the elbow trend, silhouette behaviour, and interpretability.

It should not be described as the absolute mathematically optimal value of K because the silhouette score itself is highest at K=2.

Elbow Method: Within-Cluster Sum of Squares (WCSS)



Silhouette Score vs. K (10,000 Sample Tracks)



Cluster Analysis

The final K-Means model divided the **89,741 unique tracks** into seven clusters. The clusters were analyzed using their average original audio-feature values.

Table 4. Final Cluster Profiles

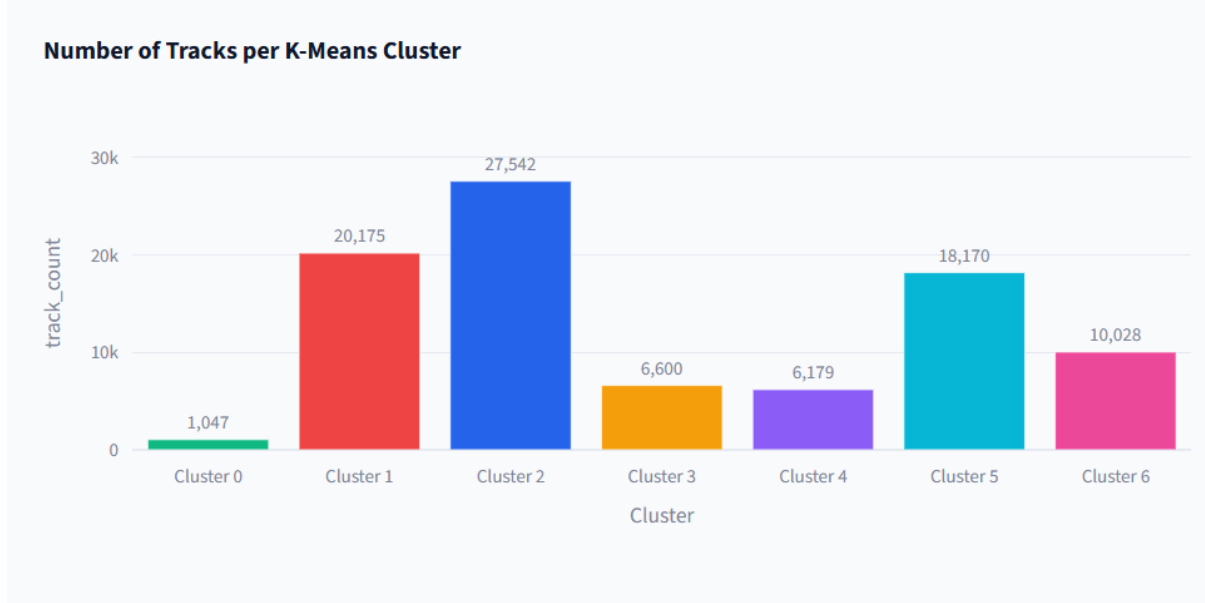
Cluster	Tracks	%	Main Characteristics
C0	1,047	1.17%	Spoken/live characteristics
C1	20,175	22.48%	High-energy, loud, rock/metal-like
C2	27,542	30.69%	Danceable, energetic, positive
C3	6,600	7.35%	Calm, acoustic, instrumental
C4	6,179	6.89%	Live/percussive characteristics
C5	18,170	20.25%	Lower-energy, acoustic/melodic
C6	10,028	11.17%	Electronic/instrumental characteristics

The largest cluster is C2, containing approximately 30.69% of the dataset. Its relatively high danceability, energy, and valence indicate an upbeat group of tracks.

Cluster C1 contains approximately 22.48% of the tracks and is characterized by high energy and loudness. Cluster C3 is considerably calmer, with high acousticness and instrumentality and low energy.

Cluster C6 also has high instrumentality and energy, suggesting a more electronic or instrumental-oriented group. C5 contains tracks with relatively lower energy and higher acousticness, representing a more melodic or relaxed profile.

These labels are descriptive interpretations based on audio features, rather than official Spotify genre classifications.



Overall Results:

The complete analysis demonstrates how PCA can simplify a relatively high-dimensional music dataset while preserving most of its information.

Table 5. Overall Project Results

Parameter	Result
Original dataset	114,000 tracks
Unique tracks after deduplication	89,741
Audio features used	9
Components retained	7
Variance retained	94.81%
Variance in PC1 + PC2	47.99%
Clustering algorithm	K-Means
K values tested	2-10
Final practical K	7
Number of final clusters	7

The preprocessing stage removed repeated track_id records before PCA. The dimensionality reduction stage then transformed the nine standardized audio features into principal components. Seven components were sufficient to retain approximately 94.81% of the variance.

The resulting PCA representation was subsequently used for K-Means clustering. This



produced seven interpretable groups with different combinations of energy, danceability, acousticness, instrumentalness, valence, and other audio characteristics.

Conclusion:

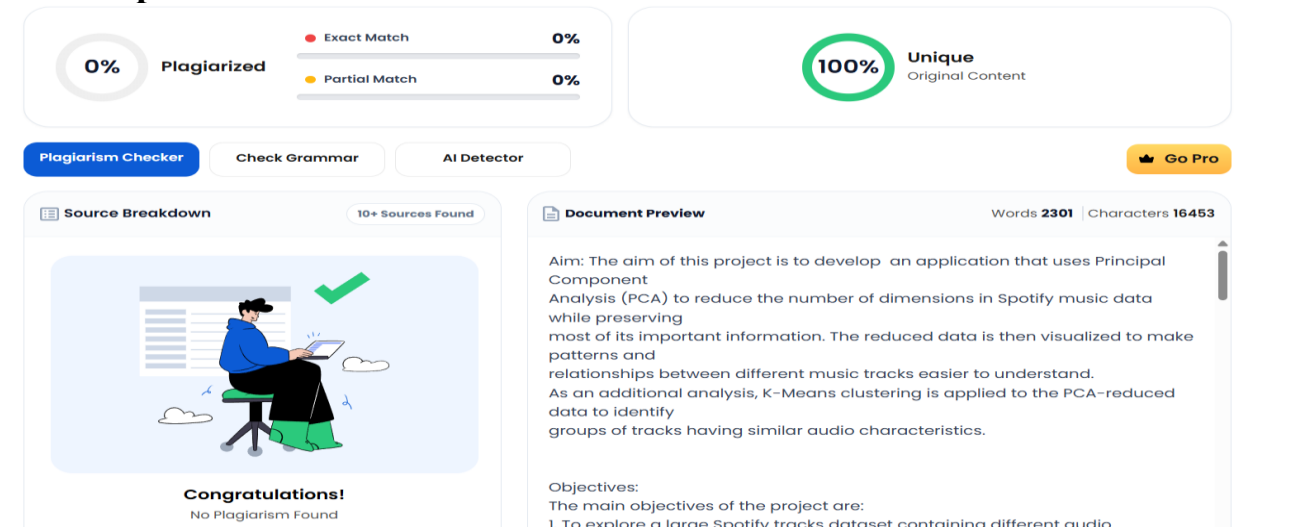
This project demonstrated the use of Principal Component Analysis (PCA) for reducing and visualizing the dimensionality of Spotify music audio features. The original dataset contained 114,000 records, which were reduced to 89,741 unique track records after track-level deduplication. Nine numerical audio features were standardized before applying PCA.

The PCA results showed that the first seven principal components retained approximately 94.81% of the total variance. This indicates that a large proportion of the information in the original nine-dimensional feature space can be represented using a smaller set of principal components. The first two components also provided a useful two-dimensional representation for visual analysis, retaining approximately 47.99% of the variance.

K-Means clustering was then applied to the seven-component PCA representation to identify groups of tracks with similar audio characteristics. Several values of K were evaluated, and K=7 was selected as a practical multi-cluster solution based on the elbow trend, silhouette behavior, and interpretability.

Overall, the project shows that PCA can be effectively combined with clustering to transform a complex music dataset into a more compact and understandable representation. The approach can help reveal broad patterns in music characteristics without depending directly on predefined genre labels.

Plagiarism Report:



Sr.No.	Roll No.	Name of student	Faculty Assessment (4M)	Report submission (3M)	Viva(3M)	Total (10M)	Student Signature
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